

Muhammad Saud Zahir

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Multan, Pakistan

Summary

Electrical Engineer with expertise in Control Systems, Signal Processing, Digital Logic Design, Machine and Deep Learning, Computational & Optimization Algorithms, Power Systems, and Numerical Simulations. Proficient in C, C++, Python, MATLAB, and leveraging programming skills for engineering challenges.

Technical Skills

Programming Languages: Python, C, C++, MATLAB.

Hardware Description Languages: Verilog

Scripting & Text Processing Languages: Perl, Tcl, Bash, AWK

Build Tools: CMake, Makefiles, Pyproject, Setuptools, Scikit-build, SIP.

Documentation Tools: Doxygen

Testing Tools: Pytest, unittest, Catch2 (C++)

Database: SQLite, PostgreSQL

Version Control: Git, GitHub, GitLab, Bitbucket

CI/CD and Agile Practices: CI/CD (GitHub Actions), Jira, Agile Methodology

Design and Modeling: OOAD, OOP, UML

C++ Libraries: Qt (GUI Framework), Eigen (Numerical Linear Algebra Library), OpenMP (Multi-processing Library), OpenGL (Specifications), fmt (C++ Format Library).

Python Libraries: TensorFlow, Sklearn, PyTorch, Pandas, NumPy, SciPy, Plotly, Flask, FastAPI, Pydantic, SQLAlchemy.

AI Models: RNN, CNN, GRU, LSTM, GNN.

Micro-controllers: Atmel 8051, Arduino UNO, ESP32

FPGA Design Tool: Intel Quartus Prime

Operating Systems: Windows & Linux

Experience

Process Automation Engineer

SOCO Engineers, Multan

August 2023 – Present

- Automation of CAE Tools (ANSA, Abaqus, OptiStruct, NASTRAN) using Python.
- Development of BMW's ShapeModule shape optimization framework based on adjoint sensitivity and gradients.
- [SIMuSPACE](#).

Internship

Multan Electric Power Company (MEPCO), Multan

July 2022 – August 2022

- Worked with the SS&T Division, Grid Stations, Operation, and T&I Division.

- Gained practical experience in grid station operations and power systems.

Education

MS Electrical Engineering

Bahauddin Zakariya University, Multan

September 2023 –

- Research: Dynamic Power Flow Simulation using Graph Neural Network.
- Major Courses: Modelling and Simulation Techniques, Optimization Algorithms, Power Transmission and Planning, Smart Grid and Communication Networks, Advanced High Voltage Engineering, Advanced Power System Protection, Advanced Techniques for Engineering Analysis, Power System Stability.

BE Electrical Engineering

School of Electrical Engineering and Computer Science (SEECS),

NUST, Islamabad

September 2019 – June 2023

- CGPA: 3.39/4
- Major Courses: Fundamentals of Programming (C), Object Oriented Programming (C++), Electronics I & II, Digital Logic Design, Signal and Systems, Communication Systems, Control Systems, Digital Signal Processing, Microprocessor Systems (x86 & Atmel 8951), Embedded Systems, Robotics I, Electromagnetic Field Theory, Microwave Engineering, Power Systems & Numerical Analysis.
- Final Year Project: Predicting Battery End of Life for Off-Grid Renewable Energy Systems.

Intermediate Pre-Engineering

Punjab Group of Colleges, Multan

2017 – 2019

Marks: 1028/1100

Major Courses: Mathematics, Physics, Chemistry.

Matriculation Science Group

Nishat High School, Multan

2015 – 2017

Marks: 1035/1100

Certifications and E-Learning

- [Machine Learning Specialization \(Stanford University\)](#)

Projects

- **Predicting Battery End of Life:** Cloud data collection with ThingSpeak, ESP32 Arduino Board, and GPR Machine Learning Model.
- **Audio Compression and Reconstruction:** MATLAB implementation.
- **Robotic Arm Control:** Based on Atmel 89C51 microcontroller.
- **Buck Converter Modelling:** MATLAB and Simulink-based analysis.

- **Eigenvalue Problem Solver:** Von Mises Iteration (Python & C++).
- **deltaFlow:** Power flow steady-state tool (C++ & FORTRAN).
- **Others:** [EELAPS](#)

Languages

English (Professional), Urdu (Native)